MIL-B-82548
28 February 1969
SUPERSEDING
MIL-B-14755(MU)
7 June 1968

MILITARY SPECIFICATION

BOMB, GENERAL PURPOSE, MARK 82, 500 POUND, EXPLOSIVE LOADING, ASSEMBLING, AND PACKING

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope - This specification contains special requirements not normally required by the applicable drawings and provides quality assurance provisions for the explosive loading, assembly, and preparation for shipment of the general purpose bomb designated as Bomb, GP, MK 82, 500 Pound.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-E-22267	Explosive Compositions, HBX Type
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-45662	Calibration System Requirements

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-129	Marking for Shipment and Storage

FSC 1325

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STANDARDS

Military (Continued)

MIL-STD-746	Radiographic Testing Requirements for Cast Explosives
MIL-STD-1167	Ammunition Data Card
MIL-STD-1168	Lot Numbering of Ammunition
MIL-STD-1314	Safety Precautions for Explosive-Loaded Items

DRAWINGS

Department of the Navy - Naval Ordnance Systems Command

DL 1380901	Bomb, MK 82 MODS, G. P. 500 lbs; Loading Assembly
LD 413540	Gages, Inspection
LD 483319	Pallet, Bomb, MK 9, MOD 0, for 500 lb. L. D. Bomb, MK 82 and MODS
1380901	Bomb, MK 82 MODS, G. P. 500 lbs; Loading Assembly
DL 2642616	Adapter, Pallet, MK 78, MOD 0

PUBLICATIONS

Department of the Army - Muntions Command

IEL-9200772	Index of Inspection Equipment Lists (MK 82, MODS, 500 Pound)

Department of the Navy - Naval Ordnance Systems Command

WR-51/1	Weapons Requirement, Truckloading, Palletized Unit Loads, Single Row Load Pattern
WR-51/6	Weapons Requirement, Truckloading Fleet Issue Unit Load, MK 36 MOD 0 for 500 lb. L. D. Bomb
WR-51/36	Weapons Requirement, Truckloading 500 lb. L. D. Bomb MK 82 and MODS, Palletized Domestic Unit Load WR-53/702 with Plastic Nose Plugs

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PUBLICATIONS

Department of the Navy - Naval Ordnance Systems Command (Continued)

WR-52/10	Weapons Requirement, Carloading 500 lb. L. D. Bomb MK 82 and MODS, Palletized Fleet Issue Unit Load MK 36 MOD 0 with Metal or Plastic Nose Plugs
WR-52/76	Weapons Requirement, Carloading 500 lb. L. D. Bomb MK 82 and MODS, Palletized Domestic Unit Load WR-53/702 with Plastic Nose Plugs
WR-52/95	Weapons Requirement, Carloading 500 Pound Low Drag Bomb MK 82 and MODS in Pallet Adapter MK 78 MOD 0 Fleet Issue Unit Load
WR-53/702	Weapons Requirement, Palletizing Domestic Unit Load 500 lb. Low Drag Bomb (G. P.) MK 82 and MODS with Metal or Plastic Nose Plugs
WR-54/31	Weapons Requirement, Palletizing Fleet Issue Unit Load MK 36 MOD 0 500 lb. L. D. Bomb
WR-54/152	Weapons Requirement, Palletizing Fleet Issue Unit Load 500 Pound Low Drag Bomb (G. P.) MK 82 and MODS with Plastic Nose Plug on Pallet Adapter MK 78 MOD 0

Department of the Air Force - OOAMA Hill Air Force Base, Utah

Transportation Packaging Order TPO 1325-294-4152	Pallet, Bomb 500 LB's L. D.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article inspection - The loaded and assembled bombs furnished under this specification shall be a product which has been inspected and passed the first article inspection specified herein. (See 4.3 and 6.1.1.)

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3.2 Conformance with documents - The bombs shall be fabricated, loaded and assembled in accordance with the drawings and applicable documents listed on DL 1380901 and shall meet all requirements of this specification.

3.3 Explosive loading - Explosive loading shall be in accordance with drawing 1380901 listed on DL 1380901.

3.4 Explosive composition - The explosive composition used in loading the assembly shall be tritonal 80/20 or H-6 explosive composition as specified by the procuring activity (see 6.1).

3.4.1 Tritonal (80/20) explosive composition - Tritonal (80/20) explosive composition shall consist of 80 percent plus or minus 3 percent trinitrotoluene (TNT) and 20 percent plus or minus 3 percent aluminum.

3.4.2 H-6 explosive composition - H-6 explosive composition shall conform to the requirements of MIL-E-22267.

3.5 Weight of applied asphaltic compound - The total weight of the asphaltic compound on the interior surfaces of the bomb assembly shall be in accordance with drawing 1380901 when tested as specified in 4.6.1.

3.6 Radiographic examination of the explosive charge - The loaded bomb shall be subjected to radiographic examination in accordance with 4.4 and 4.6.2. A shrinkage cavity will only be permitted when located 180 degrees opposite from the suspension lug holes, and more than 6 inches from the nose fuze liner, and when not in contact with the base fuze liner.

3.7 Workmanship - All interior bomb surfaces requiring application of asphaltic compound shall be free of chips, dirt, oil, grease, rust and other material before applying asphaltic compound and before loading of explosive charge. Outer bomb skin and other surfaces shall be free of chips, dirt, rust or other imperfections. All interior surfaces of the bomb assembly shall be completely coated and shall be protected from contamination. The cleaning method used to remove explosive material from the exterior surfaces of the bomb, fuze cavities, charging tubes, and threads shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning method.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved

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by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein.

4.2 Classification of inspection - The inspection of the MK 82 bomb shall be classified as follows:

(a) First article inspection (4.3)

(b) Quality conformance inspection (4.4)

4.3 First article inspection - First article inspection shall consist of all the requirements and of all the tests and inspections of this specification, and any additional inspections required by the contract or purchase order.

4.3.1 Sampling for first article inspection - Unless otherwise specified (see 6.1.1), as soon as practicable after award of a contract or order, the supplier shall furnish first article samples from each loading line for inspection to determine conformance with this specification and applicable drawings. First article samples shall consist of twenty loaded and assembled bombs which have been selected at random from a quantity of one-hundred bombs loaded and assembled in accordance with the requirements of this specification and applicable drawings. The samples shall be subjected to the first article inspection specified in 4.3. Failure of a bomb to meet all requirements shall result in rejection of the first article sample.

4.4 Quality conformance inspection - Each inspection lot shall be sampled in accordance with MIL-STD-105 and 4.5.2. Samples shall be subjected to examination of all the characteristics listed on drawing 1380901, examinations of packing characteristics (4.5.3) and tests (4.6), except the radiographic examination of the explosive charge (3.6). Upon approval of the first article samples, radiographic examination of loaded bombs shall be conducted in accordance with the following procedure: A minimum of one bomb per shift per week, per fill house shall be selected at random from a completed pallet prior to packing for shipment and tested as specified in 4.6.2 for conformance to 3.6. If a bomb fails to meet the radiographic requirements, requalification of the loading line responsible for the failure shall be required in accordance with 4.3.1. Procedures described herein shall not be resumed for the loading line until requalification testing provides evidence that control of the shrinkage cavities has been achieved. Any bomb failing the radiographic requirements shall be rejected.

4.5 Inspection provisions -

4.5.1 Inspection lot - An inspection lot shall consist of assemblies produced from one loading line in one unchanged process in accordance with the same drawing, drawing revision, specification and specification revision. Unless otherwise authorized by the procuring agency (see 6.1) the lot size shall be limited to

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maximum of one month's production. Drawing, specification or process changes not affecting safety, performance or fit, as determined by the Government, shall not necessitate changing the identification of the inspection lot. Each inspection lot shall contain explosive loaded bombs processed from batches of raw ingredients with the assurance of homogeneity within the composition limits of the applicable specification and using bomb bodies manufactured by the same manufacturer within a sixty day period. Lot numbering shall be in accordance with MIL-STD-1168.

4.5.2 Examination of characteristics - Sampling plans and procedures for the examination of critical, major, and minor characteristics shall be as specified herein. Equipment necessary for the performance of inspection shall be in accordance with 4.5.5.

4.5.2.1 Critical characteristics - Characteristics classified as "Critical" on the applicable drawings or herein shall be inspected by class on the basis of single sampling, as follows (reduced inspection is not authorized):

Lot Size	Sample	Acceptance Number
0 to 125	100 Percent	0
126 to 3200	125	0
3201 and up	Level II, MIL-STD-105	0

4.5.2.2 Major characteristics - Characteristics classified as "Major" on the applicable drawings or herein shall be inspected by class on the basis of single sampling as follows:

Sample	Acceptable Quality Level
Level II, MIL-STD-105	1.0 percent

4.5.2.3 Minor characteristics - Characteristics classified as "Minor" on the applicable drawings or herein shall be inspected by class on the basis of single sampling as follows:

Sample	Acceptable Quality Level
Level II, MIL-STD-105	2.5 percent

4.5.3 Examination of packing characteristics (see LD 483319 or DL 2642616) -

Class	Defects	Method of Inspection
Critical:	None defined	
Major:		
101	Strapping missing, improperly positioned, or broken.	Visual and Manual

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Class	Defects	Method of Inspection
102	Assembly loose	Manual
103	Pallet damaged	Visual
Minor:		
201	Evidence of poor workmanship	Visual
202	Marking missing or improperly applied	Visual

4.5.4 Process verification of explosive composition - The preparation of tritonal or H-6 explosive composition shall be subject to Government verification at the time the first batch is prepared and at random intervals during the production, but not less than once during each week of continuous operation. A record shall be maintained by the supplier for each batch prepared to insure that the components, composition, order of addition, temperature, agitation and mixing time is in accordance with mandatory procedures.

4.5.5 Inspection equipment - Unless otherwise specified by the procuring activity (see 6.1), the inspection equipment required to perform the examinations and tests prescribed in this specification shall be in accordance with the applicable departmental inspection equipment list publication IEL-9200772 or LD 413540.

4.5.6 Supply and maintenance - Supply and maintenance of the inspection equipment shall be in accordance with MIL-I-45607.

4.5.7 Calibration - The supplier shall have available a set of master gages, standards and appropriate instruments for regularly scheduled calibrations of his acceptance inspection equipment. Records of such regularly scheduled calibrations shall be maintained by the supplier and made available for review by the Government. The calibration of gages, standards and instruments shall be in accordance with MIL-C-45662.

4.6 Tests -

4.6.1 Weight of applied asphaltic compound - Prior to application of the asphaltic compound, each bomb selected in accordance with the sampling plan of 4.5.2.2 shall be weighed with an accuracy of plus or minus 1.0 pound. Record this weight as We. The application of the compound shall be in accordance with drawing 1380901. After all excess compound has been removed, weigh the bomb and record this weight as Wm. The weight of the applied asphaltic compound shall be calculated as the difference in weight, Wm-We and checked for conformance to drawing 1380901.

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4.6.2 Radiographic examination of the explosive charge - Each bomb selected shall be radiographed in two planes in accordance with the requirements of MIL-STD-746. One plane shall be through the lug and bomb centers and the other at 90 degrees to it. Shrinkage cavities if any, shall be checked for conformance to 3.6.

5. PREPARATION FOR DELIVERY

5.1 Levels A, B and C packaging - Not applicable.

5.2 Packing -

5.2.1 Level A - For Navy shipments, the loaded and assembled bombs shall be packed on pallet adapters LD 483319 or DL 2642616 in accordance with Naval Weapons Requirement WR-54/31 or WR-54/152 respectively. Loaded and assembled bombs shall be packed in accordance with Transportation Packaging Order (TPO) drawing 1325-294-4152 for Air Force shipments.

5.2.2 Level B - (Applicable for Air Force shipments only). The loaded and assembled bombs shall be packed in accordance with Transportation Packaging Order (TPO) drawing 1325-294-4152.

5.2.3 Level C - The loaded and assembled bombs shall be packed in accordance with Naval Weapons Requirement WR-53/702 for Navy shipments and in accordance with Transportation Packaging Order (TPO) drawing 1325-294-4152 for Air Force shipments.

5.3 Marking - Each bomb shall be marked as specified on drawing 1380901. Pallets shall be marked for shipment in accordance with MIL-STD-129.

5.3.1 Ammunition data cards - When specified (see 6.1), data cards shall be prepared in accordance with MIL-STD-1167 except that only space numbers 1 through 14, 23, and 26 shall be completed. Space number 23, identification of components, shall be limited to the inert bomb body.

5.4 Delivery -

5.4.1 Level A -

5.4.1.1 Carloading - The bombs packed for Navy shipments as specified in 5.2.1 shall be loaded in accordance with Naval Weapons Requirement WR-52/10 or WR-52/95. Bombs packed for Air Force shipments as specified in 5.2.1 shall be loaded in accordance with Naval Weapons Requirement WR-52/76.

5.4.1.2 Truckloading - The bombs packed for Navy shipments as specified in 5.2.1 shall be loaded in accordance with Naval Weapons Requirement WR-51/6 associated with WR-54/31, and WR-51/1 associated with WR-54/152. Bombs packed for Air Force shipments as specified in 5.2.1 shall be loaded in accordance with Naval Weapons Requirement WR-51/36.

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5.4.2 Level B -

5.4.2.1 Carloading - Bombs packed for Air Force shipments as specified in 5.2.2 shall be loaded in accordance with Naval Weapons Requirement WR-52/76.

5.4.2.2 Truckloading - Bombs packed for Air Force shipments as specified in 5.2.2 shall be loaded in accordance with Naval Weapons Requirement WR-51/36.

5.4.3 Level C -

5.4.3.1 Carloading - The bombs packed as specified in 5.2.3 for Navy and Air Force shipment shall be loaded in accordance with Naval Weapons Requirement WR-52/76.

5.4.3.2 Truckloading - The bombs packed as specified in 5.2.3 for Navy and Air Force shipment shall be loaded in accordance with Naval Weapons Requirement WR-51/36.

6. NOTES

6.1 Ordering data - Procurement documents should specify the following:

(a) Title, number and date of this specification

(b) Explosive composition required (see 3.4 through 3.4.2)

(c) Quantity of bombs

(d) When lot size requirements are necessary other than specified (see 4.5.1)

(e) Level of packing and delivery required (see 5.2 and 5.4)

(f) Applicable inspection equipment list publication (see 4.5.5)

(g) When supplier designed inspection equipment and inspection equipment data lists are required (see 4.5.5 and 6.3)

(h) Additional safety precautions as necessary (see 6.2)

(i) Data cards when required (see 5.3.1)

6.1.1 Contracts or orders should specify the following provisions for first article inspection.

6.1.1.1 Whether first article inspection is required - When a contractor is in continuous production from contract to contract, consideration should be given to waive the first article inspection. If first article inspection is required, indicate:

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(a) Where the first article inspection is to be conducted (at the supplier's or Government plant).

(b) That approval of first article samples or the waiving of the first article inspection shall not relieve the supplier of his obligation to fulfill all other requirements of this specification and contract.

6.2 General safety requirements - The preparation and handling of the items covered by this specification, and the subassemblies thereof involve hazardous operations and therefore require explosives safety precautions. Use of this specification will not be construed as to relieve the supplier or manufacturer of responsibility for the safety of his operation. The provisions of MIL-STD-1314 shall apply for all Government, suppliers, or manufacturers loading plants preparing and handling the items covered by the specification. When specified by the procuring agency, additional safety requirements will be specified as deemed necessary (see 6.1) in the fulfillment of the contract.

6.3 Inspection equipment - When specified by the procuring agency (see 4.5.5 and 6.1), supplier designed inspection equipment shall be in accordance with 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists - Inspection equipment lists shall be prepared by the supplier, in accordance with MIL-D-45608, indicating the availability of inspection equipment designs, the numbers of drawings or Federal Stock Numbers of existing equipment designs or codes as indicated in 6.3.2. Design action required of the supplier is described in 6.3.3 and 6.3.4. The supplier will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Supplier Design". These supplier designs shall be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes - The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated by the following codes:

CD - Supplier design on controlled supplier format and/or commercial equipment.

DOD - Military design, mandatory for use.

6.3.3 Military designs - Military designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The supplier need provide no design when a military design is listed for an item of inspection equipment. Military designs fall into two basic classifications; mandatory (designated "DOD") and non-mandatory (indicated by drawing or

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Federal Stock Number). When an inspection equipment list references mandatory military designs, the supplier shall comply with, and use these designs accordingly. The supplier may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such supplier prepared designs or commercial equipment selections shall be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Supplier designs - Supplier designs are designs of inspection equipment for which the Government has assigned design responsibility to the supplier. Supplier designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating, and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, supplier designs may be developed on the format the supplier normally employs in his equipment design procedures provided such format reflects the detail and information specified above, subject to the following controls: All submitted supplier designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Suppliers shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of supplier designs - Unless otherwise specified by this specification, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the departmental agency as designated by the contracting officer. All other designs of inspection equipment shall be approved by the Government agency administering the contract as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Design review by the military should normally be accomplished within one month after receipt.

Custodian:
Army - MU
Navy - OS
Air Force - 70

Preparing activity:
Navy - OS
Project No. 1325-0123

Review interest:
Army - MU
Navy - AS, OS
Air Force - 70, 18

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof).*

SPECIFICATION	
MIL-B-82548	BOMB, GENERAL PURPOSE, MARK 82, 500 POUND, EXPLOSIVE LOADING, ASSEMBLING, AND PACKING

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED *(Printed or typed name and activity)*	DATE

DD FORM 1 NOV 64 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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